Deduce the World: How Far Can Reasoning Go?

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Results & figures

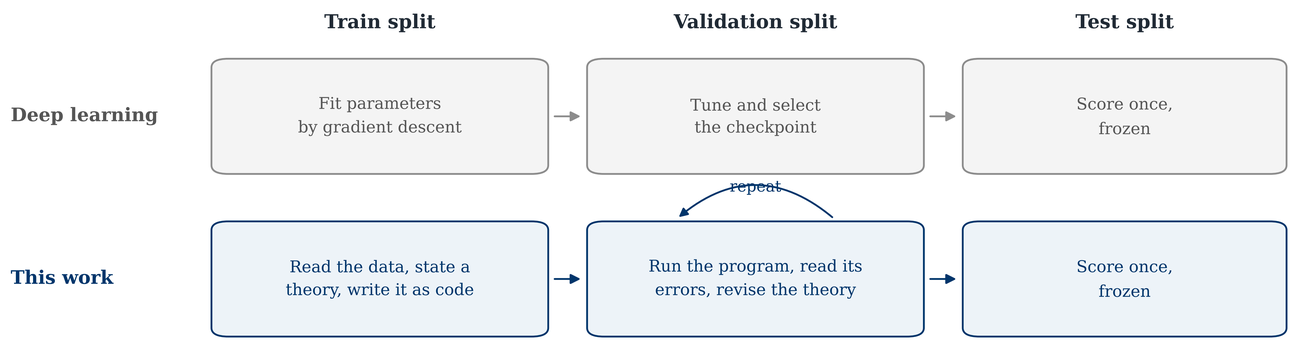
1 Theories, not fitted predictors

Science deduces the world from first principles, revising its theories where their predictions fail.

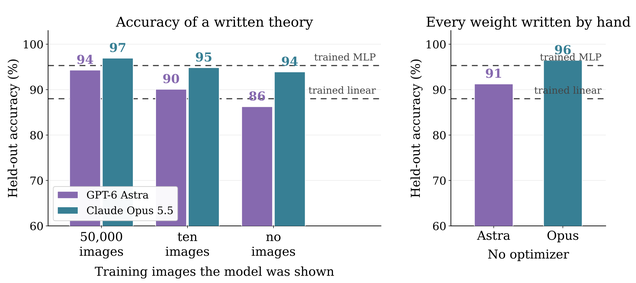
We ask whether a reasoning model can form scientific theories, and whether those theories generalize better than deep learning.

May read papers and the web, measure the training data, inspect its own errors

May not fit parameters to labels, search automatically over rules, store examples, see the test split



2 How much data a theory needs

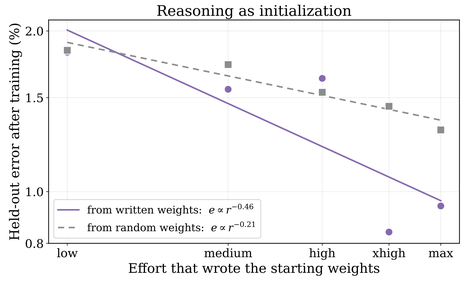


MNIST, 60,000 training and 10,000 test images. Best of five reasoning efforts; dashed lines are trained models.

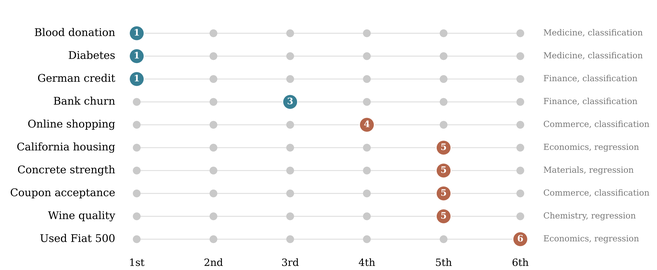
> Given no training images at all, a theory of how digits are written classifies most of them.

Reasoning as initialization

Networks initialized from written weights train to lower error, and the gap widens with effort.



3 The same protocol, ten fields

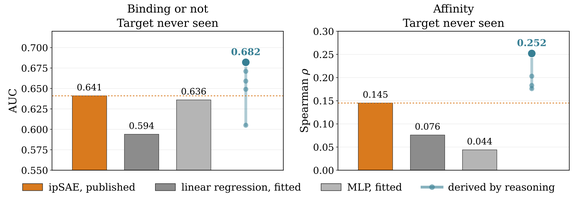


Ten tasks from TabArena, with fixed splits. Each ranks the theory against five tuned models on the same rows.

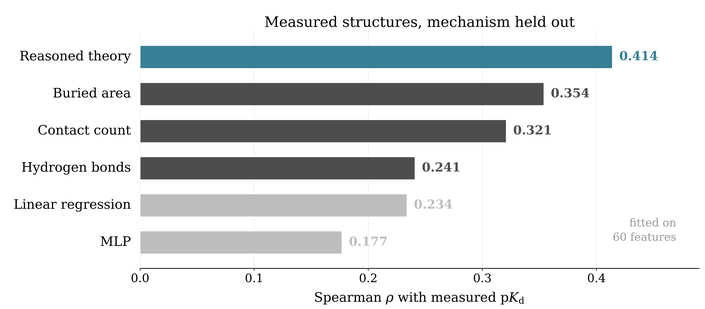
> Theories win on small clinical and credit tables, and trail boosted trees where the signal lies in feature interactions.

4 Deriving a scientific score

Each session receives a structure predictor's error matrix and nothing else.



704 designed binders, holding out a whole target.



Measured structures, one mechanism of association held out whole.

> A rule derived from physics beats the published score on a target it has never seen, and every unfitted physical quantity beats both fitted models.

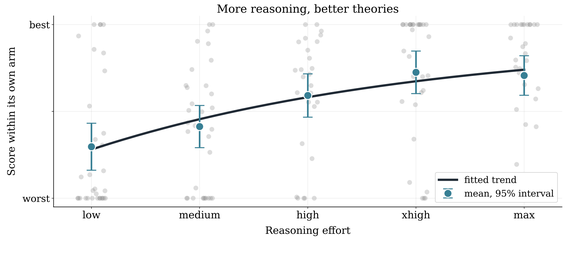
The theory it stated

Harmonic well	Equipartition sets the force constant to the thermal energy over the square of the fluctuation.
Stiffness is depth	Curvature at the minimum is proportional to depth.
Divide out the predictor	Measuring the interface error against the binder's own internal error removes predictor quality, and the units.
Count what binding confines	Association removes six rigid-body degrees of freedom, each contributing half the thermal energy times the log of the stiffness.

$$\kappa = \left(\frac{\sigma_0}{\sigma_{\text{iface}}}\right)^2, \qquad \mathrm{p}K_{\mathrm{d}} = \mathrm{p}K_{\mathrm{d,ref}} + 3\log_{10}\kappa.$$

The three is six halved. Nothing is fitted.

5 Scaling reasoning effort



Every arm on one axis, normalized to its own range.

6 So how far can reasoning go?

- Deep learning dominates the domains that are data-rich and theory-poor.
- Those are exactly the domains whose first principles nobody has written down yet.
- We hypothesize that as reasoning improves, models will supply them, and stated theories will replace fitted predictors.

7 What this does not show

- These models have read the literature of every domain we test, so recall and derivation are not fully separable.
- The protein cohort is curated candidates, held-out counts are small, and nothing is bootstrapped.
- The initialization result is one model family at two seeds, and an effort setting is not a measure of compute.